



CYBER DEFENSE INTELLIGENCE // M.C.A
DISSERTATION

Unified Multi-Environment Cyber Threat Detection & Response Platform

Autonomous Telemetry Ingestion • MITRE ATT&CK Kill-Chain
• ML Anomaly Engine • SHA-256 Audit Ledger

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Objective

To engineer an autonomous, multi-environment cyber threat intelligence and response platform that eliminates enterprise alert fatigue by consolidating real-time log ingestion, dual-engine ML anomaly detection, MITRE ATT&CK kill-chain correlation, human-in-the-loop response orchestration, and a cryptographic SHA-256 tamper-evident audit ledger.



About Project

Unified Threat Platform is a full-stack cybersecurity SOC operations platform designed to detect complex multi-stage intrusions across on-prem, cloud, and DMZ infrastructures. It unifies Elastic Common Schema (ECS) normalization, unsupervised ML anomaly detection (Isolation Forest), automated containment playbooks with dual-authorization approval gates, and mathematical proof-of-integrity in a unified browser command center.





Platform Key
Features



Multi-
Environment:
Monitors Corp,
Cloud, and DMZ
perimeters in a
single pane.



Dual Detection
Engine: Fast



Student Operator & Output Flow



SECURITY
ANALYST



Monitor Live SOC



Triage Kill-Chain



Tools & Technologies
Used

DEVELOPMENT &
DEPLOYMENT TOOLS



VS
Code



Git &
GitHub



Vercel
Cloud



Docker &
Compose

file:///C:/Users/User/OneDrive/Desktop/always%20gratitude/project_flash_card.html

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YAML rules +
Unsupervised
ML Isolation
Forest.

✓ **MITRE Kill-Chain:** Real-time mapping across all 7 ATT&CK lifecycle stages.

✓ **Entity Clustering:** Drops alert fatigue by 85% through graph aggregation.

✓ **Dual-Auth Approval Gate:** Human-in-the-loop protection for high-risk actions.

✓ **SHA-256 Hash Chain:** Cryptographic tamper-evident audit trail & evidence lock.

✓ **Real-Time WebSockets:** Sub-second telemetry mesh and interactive charts.

✓ **Enterprise IAM & MFA:** RBAC security with TOTP one-time passcode support.

Review ML Anomaly

Approval Gate

Verify Hashes

Export Audit Docs

⚡ **VISUAL OUTPUT VIEW FLOW**

Raw Logs → Normalized → Threat Alert → Incident Story → SHA-256 Proof

pytest (53 Passed)

Vite v5.4

⚙️ **CORE ARCHITECTURE & LIBRARIES**

React 18 & TS

FastAPI & Python

Scikit-Learn (ML)

Tailwind CSS

SQLite / Postgres

Redis & Celery

SHA-256 Ledger

MITRE ATT&CK

 **Case Study Comparison**
(ApexFin Corp Enterprise APT Breach)

✗ **Legacy SOC in Case Study:** 1,400+ disconnected alerts flooded analysts; 7.5 hours MTTR; manual emails delayed response; flat logs lacked tamper verification.

✓ **Unified Threat Platform:** Collapsed alerts into 1 unified incident story; automated IP block in < 2 minutes; 100% cryptographic SHA-256 evidence integrity.

 **PROJECT LINKS**

GitHub: github.com/Pravallika2025/unified-threat-platform

Vercel Live: frontend-phi-indo1-81.vercel.app

Swagger Docs: frontend-phi-indo1-81.vercel.app/docs

CDN SPA: pravallika2025.github.io/unified-threat-platform/